

Day-Ahead Electricity Price Forecasting in the Indian Energy Exchange: Gradient-Boosted Ensembles at 15-Minute Resolution

Uttam Paliwal
uttam232002@gmail.com

August 22, 2026

Abstract

We present a systematic empirical study of machine learning models for day-ahead electricity price forecasting in the Day-Ahead Market (DAM) of the Indian Energy Exchange (IEX), evaluated at the exchange’s native 15-minute resolution of 96 clearing blocks per day. Using approximately six and a half years of block-level market data (January 2020–May 2026), comprising roughly 2.2 million observations scraped from public exchange records, we benchmark five model families—seasonal-naive persistence, ridge regression, random forests, XGBoost, and LightGBM—under a strict temporal evaluation protocol in which all models are trained on January 2020–December 2024 and evaluated once on an untouched 17-month holdout (January 2025–May 2026). Gradient-boosted ensembles dominate: XGBoost attains $R^2 = 0.867$ with RMSE of 966 Rs/MWh and a weighted absolute percentage error (WAPE) of 13.8%, against $R^2 = 0.555$ and RMSE of 1701 Rs/MWh for a strong weekly persistence baseline—a 43% reduction in root-mean-squared error and a 56% relative improvement in R^2 . A paired ablation shows that adding hourly apparent temperatures for Delhi and Mumbai changes holdout R^2 by at most ± 0.004 , indicating that calendar structure and autoregressive lags capture essentially all exploitable signal at this horizon. We additionally evaluate directional skill via a sign-classification protocol (AUC-ROC up to 0.792; 80.2% direction accuracy for the champion model) and document stable monthly error over the full 17-month holdout. The full pipeline—data collection, feature engineering, training, benchmarking, and dashboards—is released as open source.

Keywords: electricity price forecasting; day-ahead market; Indian Energy Exchange; gradient boosting; XGBoost; LightGBM; time-series evaluation

Code and pipeline: <https://github.com/uttampaliwal/power-price-predictor>. Market data are publicly available from the Indian Energy Exchange; weather data from Open-Meteo.

1 Introduction

Electricity is among the most volatile commodities traded on organized markets: it is largely non-storable at scale, demand is weather- and activity-driven, and prices must clear simultaneously across fine-grained delivery intervals. In India’s Day-Ahead Market (DAM), operated by the Indian Energy Exchange (IEX), buyers and sellers submit block-wise bids in a double-sided closed auction, and a single Market Clearing Price (MCP) is published for each of the 96 fifteen-minute blocks of the following day [Indian Energy Exchange Ltd. \(2026\)](#). Within a single day, DAM prices routinely

traverse an order of magnitude, from a few hundred Rs/MWh overnight to several thousand during evening peaks, subject to a regulatory price ceiling of Rs10,000/MWh [Central Electricity Regulatory Commission, Government of India \(2024\)](#).

Accurate day-ahead price forecasts are operationally valuable to every class of market participant. Generating stations use them to schedule maintenance and optimize which plants to offer into the exchange; large industrial consumers and open-access buyers use them to shift load into cheap blocks; and traders and distribution companies (DISCOMs) use them to construct bids that balance cost against procurement risk. Forecast errors translate directly into scheduling mismatches, imbalance exposure, and lost arbitrage.

Despite India operating one of the world’s largest real-time power markets by volume, the electricity price forecasting (EPF) literature remains heavily concentrated on European and North American markets [Weron \(2014\)](#); [Lago et al. \(2021\)](#). Published benchmarks rarely examine sub-hourly resolution for Indian data, and practitioner-oriented studies that combine rigorous temporal validation with deployment-ready engineering are scarce.

This paper makes the following contributions:

1. **A reproducible benchmark on long-horizon Indian market data.** We curate approximately 2.2 million block-level DAM observations spanning January 2020–May 2026, collected directly from public IEX records, and evaluate five model families under a strict temporal split: training on 2020–2024, model selection only through end-2024, and a single untouched evaluation over a 17-month holdout (January 2025–May 2026). To our knowledge this holdout horizon is substantially longer than what is typical in EPF studies, where test windows of weeks to months are common [Lago et al. \(2021\)](#).
2. **A dual regression-plus-direction evaluation protocol.** Beyond standard regression metrics (R^2 , RMSE, MAE, MAPE, WAPE), we formalize a directional task—did the price rise by at least 1% versus yesterday’s same block?—evaluated with AUC-ROC and F1, reflecting how trading desks actually consume forecasts.
3. **A weather-ablation finding.** Adding hourly apparent temperatures for Delhi and Mumbai to an already strong feature set shifts holdout R^2 by no more than ± 0.004 across four model families, empirically demonstrating that for day-ahead auctions—where participants’ bids already embed weather expectations—raw meteorological regressors add essentially no incremental value.
4. **Evidence of long-horizon stability.** Monthly RMSE over the 17-month holdout remains within a narrow band for all learned models while the naive baseline degrades sharply during high-volatility spring months, quantifying robustness to regime drift.
5. **An open, deployment-grade system.** The complete pipeline—daily scraping, leakage-safe feature construction, per-model training with early stopping, benchmarking, drift monitoring, and interactive dashboards—is released as open source [Paliwal \(2026\)](#).

Our headline result is that gradient-boosted tree ensembles (XGBoost, LightGBM) explain 86.7% of holdout price variance at 15-minute granularity, reducing RMSE by 43% relative to a weekly persistence baseline, with a weighted absolute error of 13.8%. Equally instructive is what does *not* work: weather features are nearly inert, and the linear model that is weakest at level prediction (ridge regression) is the strongest at ranking price direction.

The remainder of the paper is organized as follows. Section 2 situates our study in the EPF literature. Section 3 describes the market context, data collection, and dataset. Section 4 details

features, models, and the evaluation protocol. Section 5 presents benchmark, ablation, stability, and direction-classification results. Section 6 analyzes the findings, Section 7 discusses limitations, and Section 8 concludes.

2 Related Work

Weron (2014) provides the canonical survey of electricity price forecasting, cataloguing statistical (ARIMA-type, GARCH-type, jump-diffusion), machine-learning (neural networks), and hybrid approaches, and highlighting persistent methodological pitfalls: short test windows, inconsistent metrics, and accidental data leakage. More recently, Lago et al. (2021) established an open-access benchmark for day-ahead forecasting across multiple European markets and found that leak-free, like-for-like comparisons compress much of the reported spread between sophisticated methods, with carefully tuned statistical and machine-learning models performing comparably. Our protocol follows their best practices—chronological splits, multi-metric reporting, and honest baselines—applied to an understudied market.

Probabilistic extensions are surveyed by Nowotarski and Weron (2018); we restrict attention to point forecasts but identify probabilistic outputs as priority future work.

On the model side, gradient-boosted decision trees (Friedman, 2001) have become dominant tabular learners: XGBoost (Chen and Guestrin, 2016) introduced regularized second-order boosting with sparsity-aware splits, and LightGBM (Ke et al., 2017) accelerated training via histogram-based splitting and leaf-wise growth. Random forests (Breiman, 2001) remain a robust bagging baseline. Among deep sequence models, LSTM networks (Hochreiter and Schmidhuber, 1997) and attention-based architectures such as the Temporal Fusion Transformer (Lim et al., 2021) target multi-horizon forecasting with rich covariates; Prophet (Taylor and Letham, 2018) targets interpretable decomposable trends. Consistent with findings in Lago et al. (2021), our results show that on leakage-safe tabular features, boosted trees set a very strong bar.

Feature importance in tree ensembles can be interpreted via gain-based rankings or Shapley values (Lundberg and Lee, 2017); we report gain-based importances and qualitative driver analysis.

For error measurement, MAPE is well known to be unstable when true values approach zero (Hyndman and Koehler, 2006); because capped and near-floor blocks occur in the DAM, we adopt WAPE (also called WMAPE), which normalizes total absolute error by total realized volume-weighted price, alongside MAPE for comparability.

Finally, market context matters: IEX runs a uniform-price double auction cleared separately for each 15-minute block, with bid books closing the day before delivery. This institutional structure motivates our central hypothesis that lagged prices and calendar effects—rather than exogenous weather—carry most of the forecastable signal, a hypothesis our ablation confirms.

3 Market Context and Data

3.1 The IEX Day-Ahead Market

The Indian Energy Exchange operates India’s principal spot platform for electricity. In the DAM:

- Each trading day is divided into 96 fifteen-minute *time blocks*.
- Participants submit purchase and sell bids (volume in MW, with price limits) per block before gate closure on the previous day.

Table 1: Dataset summary. Block counts refer to 15-minute DAM observations after removing rows with missing MCP.

Property	Value
Market / instrument	IEX Day-Ahead Market, uniform-price MCP
Resolution	15 minutes (96 blocks/day)
Full span	Jan 2020 – May 2026 (≈ 6.5 years)
Training period	Jan 2020 – Dec 2024 (≈ 1.75 M rows)
Internal validation	Last 60 days of 2024 (Nov–Dec 2024)
Holdout (untouched)	Jan 2025 – May 2026 (≈ 480 K rows)
Per-row fields	date, time block, purchase bid, sell bid, MCV, scheduled volume, MCP
Weather covariates	hourly apparent temperature, Delhi and Mumbai
Price regime	typically 10^2 – 10^4 Rs/MWh; regulatory ceiling Rs10,000/MWh

- A market-clearing algorithm computes, per block, the Market Clearing Volume (MCV) and a uniform Market Clearing Price (MCP).
- Prices are quoted in Rs/MWh and bounded above by a regulatory ceiling of Rs10,000/MWh ([Central Electricity Regulatory Commission, Government of India, 2024](#)); in stressed periods (heat waves, supply shortages) prices frequently pin near this cap.

Two structural properties shape the forecasting problem. First, *day-ahead closure*: all bids, which aggregate participants’ own weather and demand intelligence, are locked in before delivery day, so the forecaster must predict prices conditional on information available one day ahead. Second, *block-level homogeneity*: within a day, adjacent blocks exhibit strong autocorrelation, while the same block across days exhibits pronounced weekly seasonality driven by industrial activity patterns.

3.2 Data Collection and Dataset

We collected the full public DAM snapshot table (date, time block, purchase bid MW, sell bid MW, MCV MW, final scheduled volume MW, MCP Rs/MWh) day-by-day from the IEX market-data portal [Indian Energy Exchange Ltd. \(2026\)](#), covering January 1, 2020 through May 2026. Table 1 summarizes the resulting corpus. Hourly historical apparent temperatures for Delhi (28.61°N, 77.21°E) and Mumbai (19.08°N, 72.88°E)—the load centers of India’s northern and western regions—were obtained from the Open-Meteo historical archive [Zippenfenig \(2026\)](#).

Three data-quality considerations were addressed during ingestion: (i) exchange-side summary rows (averages, maxima) were filtered out so that only true 15-minute blocks remain; (ii) numeric fields containing thousands separators were normalized; and (iii) days with partial or missing tables were retained as gaps rather than imputed at the raw stage, with missingness handled downstream in feature construction.

4 Methodology

4.1 Problem Formulation

Let $y(d, b)$ denote the MCP (Rs/MWh) for day d and block index $b \in \{0, \dots, 95\}$, where $b = 0$ corresponds to 00:00–00:15 IST. The task is day-ahead point forecasting: using only information available up to the end of day $d - 1$, predict the entire vector $(y(d, 0), \dots, y(d, 95))$. We learn a

single global function $\hat{y}$ over all blocks, feeding intra-day position as explicit features, rather than fitting 96 separate models; this shares statistical strength across blocks and simplifies deployment.

4.2 Feature Engineering

All features are constructed strictly from information available before delivery day; rolling statistics are computed with a one-day shift to prevent leakage. The feature set comprises 18 variables in four groups:

- **Calendar/time (10):** block index b ; hour; day-of-week; day-of-year; month; year; weekend indicator; Indian public-holiday indicator; a four-way season code (winter Dec–Feb, summer Mar–May, monsoon Jun–Sep, post-monsoon Oct–Nov); and a five-way hour-bucket code separating off-peak night, morning peak (06–10), afternoon (10–14), pre-evening (14–18), and evening peak (18–22).
- **Autoregressive lags (2):** $y(d - 1, b)$ and $y(d - 7, b)$, i.e., the same block yesterday and last week.
- **Rolling statistics (4):** mean and standard deviation of $y(\cdot, b)$ over trailing 7- and 30-day windows, computed per block with a one-day shift.
- **Weather (2):** hourly apparent temperature for Delhi and Mumbai, merged onto each observation by timestamp.

Rows whose lag/rolling features fall inside the first seven days of history are excluded from model fitting. Weather covariates are forward/backward filled across occasional hourly gaps.

4.3 Models

We compare five families spanning the accuracy–complexity spectrum:

1. **Seasonal-naïve persistence:** $\hat{y}(d, b) = y(d - 7, b)$. This is a demanding baseline in strongly weekly data.
2. **Ridge regression:** standardized features with L2 regularization ($\alpha = 1$); a fast, interpretable linear reference.
3. **Random forest:** 300 fully grown bagged trees (Breiman, 2001).
4. **XGBoost** (Chen and Guestrin, 2016): depth-7 boosted trees, learning rate 0.05, subsample and column subsample 0.8, minimum child weight 5, L2/L1 regularization 1.0/0.1, up to 1000 rounds with early stopping (patience 50) on the time-ordered validation set.
5. **LightGBM** (Ke et al., 2017): leaf-wise growth with 63 leaves, depth 7, learning rate 0.05, minimum child samples 20, otherwise identical regularization and early-stopping configuration.

Additionally, the released pipeline implements a sequence model—a two-layer LSTM (Hochreiter and Schmidhuber, 1997) consuming 7-day input sequences (672 steps) and emitting all 96 next-day blocks jointly—as a foundation for future deep-learning comparisons; detailed benchmarking of sequence architectures (including TFT-style attention models (Lim et al., 2021)) is left to subsequent work, and no sequence-model numbers are claimed here.

4.4 Training and Evaluation Protocol

We enforce a strict chronology at three levels:

- **Train/validation/test separation.** Models train on January 2020–October 2024; the final 60 days of 2024 serve as the early-stopping and model-selection validation window; the holdout from January 2025 onward is scored exactly once, with no tuning, re-fitting, or threshold adjustment performed on it.
- **Leakage-safe features.** All lag/rolling features use shifted windows (Section 4.2); normalization statistics for the LSTM derive from training data only.
- **No shuffling.** Samples are never randomly permuted across split boundaries; random splits are known to inflate time-series scores severely (Weron, 2014; Lago et al., 2021).

All experiments fix random seeds and run on identical feature matrices, so cross-model differences are attributable to the learners alone.

4.5 Metrics

Regression. For held-out pairs $(y_i, \hat{y}_i)$ we report RMSE, MAE, MAPE, the coefficient of determination R^2 , and WAPE:

$$\text{WAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|} \times 100. \quad (1)$$

WAPE is preferred over MAPE as the operational accuracy measure because MAPE explodes on blocks whose realized price approaches zero (or pins at the regulatory floor), a recurring event in the DAM (Hyndman and Koehler, 2006).

Direction classification. For trading decisions the sign of the day-over-day move often matters more than its magnitude. Define the binary label $z_i = \mathbb{1} \left[\frac{y_i - y_{i-96}}{|y_{i-96}|} \geq 0.01 \right]$, i.e., price rose by at least 1% versus yesterday’s same block. We compute AUC-ROC between realized labels z and labels derived from predicted prices, plus F1 of the binarized predictions. This dual protocol separates *level* skill (operations, scheduling) from *directional* skill (trading).

5 Results

5.1 Main Benchmark

Table 2 reports holdout performance over January 2025–May 2026; Figure 1 visualizes the comparison across six metrics.

Three observations stand out. First, gradient boosting dominates level accuracy: XGBoost achieves $\text{RMSE} = 966 \text{ Rs/MWh}$ versus 1701 for weekly persistence, a **43% error reduction**, and raises R^2 from 0.555 to 0.867—a 56% relative improvement, equivalently reducing unexplained variance by roughly 70%. Second, the gap between XGBoost and LightGBM is negligible (RMSE differs by 2 Rs/MWh), consistent with both implementing the same additive-tree functional form with different optimization strategies. Third, all four learned models beat the naive baseline decisively, confirming the features carry genuine signal beyond weekly autocorrelation.

In absolute terms the champion’s 13.8% WAPE means the average block-level forecast is within roughly one-seventh of the realized price—operationally meaningful given that day-over-day DAM prices routinely move by far more than this.

Table 2: Holdout benchmark (January 2025–May 2026, $\approx 480\text{K}$ block-level predictions). Best value per column in bold. WAPE is the primary operational error metric; MAPE is shown for comparability with prior literature. Direction metrics use the $\geq 1\%$ day-over-day rise label.

Model	RMSE	WAPE %	MAPE %	R^2	AUC-ROC	F1
XGBoost	966	13.8	29.5	0.867	0.779	0.681
LightGBM	968	13.8	30.5	0.867	0.776	0.677
Random Forest	987	14.0	27.4	0.862	0.742	0.629
Ridge	1136	16.4	29.1	0.817	0.792	0.623
Naive (lag-7d)	1701	23.4	35.9	0.555	0.734	0.698

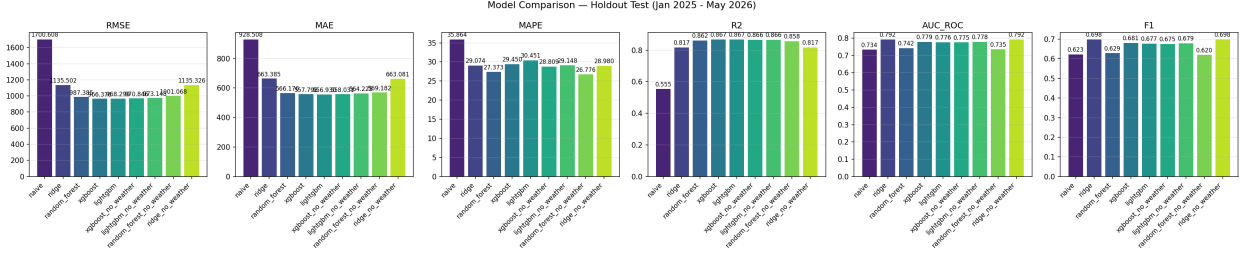


Figure 1: Holdout model comparison (Jan 2025–May 2026) across RMSE, MAE, MAPE, R^2 , AUC-ROC, and F1. Boosted-tree models (xgboost, lightgbm) and their no-weather variants cluster tightly at the best regression performance; ridge leads AUC-ROC. No-weather variants track their weather counterparts closely throughout.

5.2 Weather Ablation

To isolate the marginal value of meteorological covariates, each model family was trained twice on identical hyperparameters and identical splits: once with the two apparent-temperature series, once without. Table 3 shows the holdout deltas.

The effect is uniformly negligible. The largest change in either direction is $|\Delta R^2| \leq 0.004$. We attribute this to the institutional structure of the day-ahead auction (Section 3.1): by the time the forecaster acts, every participant’s bid already aggregates sophisticated private weather and demand intelligence, so raw city-level temperatures are redundant with respect to the lagged-price signal. This finding carries practical weight—two-point temperature measurements are a tempting but unproductive feature category here—and aligns with the broader lesson that forecast value must be demonstrated against leakage-safe baselines rather than asserted (Lago et al., 2021).

5.3 Temporal Stability Over 17 Unseen Months

Figure 2 plots monthly RMSE across the holdout. All learned models remain tightly clustered (roughly 600–1400 Rs/MWh month-to-month), with no upward trend indicative of decay over 17 months without retraining. The naive baseline, by contrast, degrades severely during high-volatility months—exceeding 2300–2600 Rs/MWh during March–June 2025 and again in spring 2026—precisely when heat-wave-driven demand surges break the previous week’s pattern. Learned models absorb these episodes far more gracefully, though their own error also rises in those months, tracking volatility rather than drifting with time.

This has a direct governance implication for production systems: error escalation in electricity

Table 3: Weather ablation on the holdout. $\Delta R^2 = R^2_{\text{no wx}} - R^2_{\text{wx}}$; negative values mean removing weather *improved* the model.

Model	R^2 (wx)	R^2 (no wx)	ΔR^2	AUC (wx $\rightarrow$ no wx)
XGBoost	0.867	0.866	+0.001	0.779 $\rightarrow$ 0.775
LightGBM	0.867	0.866	+0.001	0.776 $\rightarrow$ 0.778
Random Forest	0.862	0.858	+0.004	0.742 $\rightarrow$ 0.735
Ridge	0.817	0.817	0.000	0.792 $\rightarrow$ 0.792

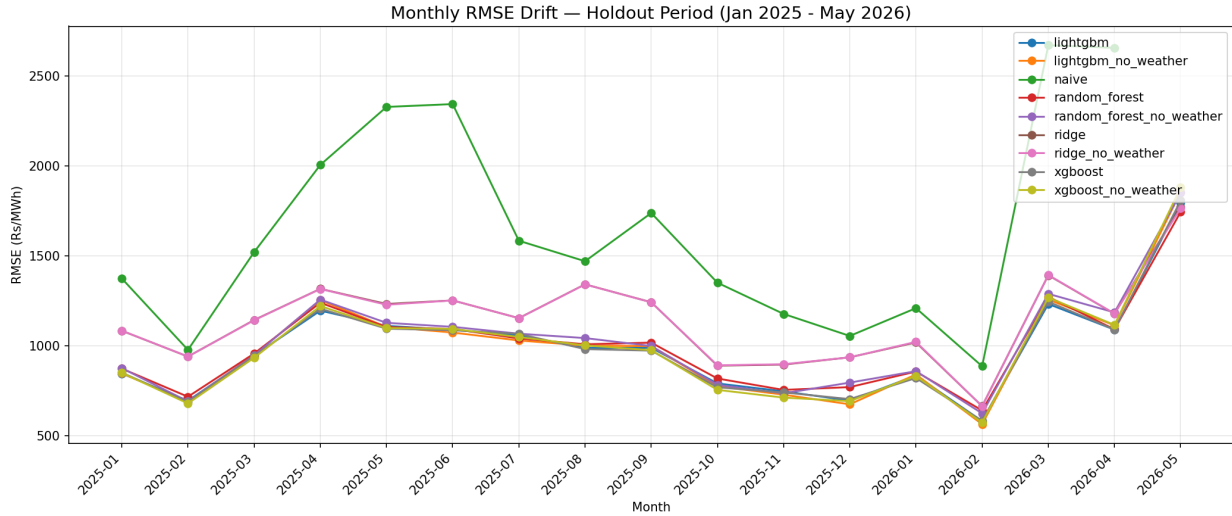


Figure 2: Monthly RMSE drift over the 17-month holdout (January 2025–May 2026). Learned models (clustered lines, lower band) remain stable while the naive weekly-persistence baseline (green) spikes during high-volatility spring months. Weather and no-weather variants of each model are visually indistinguishable, echoing Table 3.

markets should first be diagnosed as volatility regime shift, not model staleness; the naive baseline serves as a cheap canary distinguishing the two.

5.4 Direction Classification

The rightmost columns of Table 2 reveal a trade-off invisible to regression metrics. Ridge regression— weakest of the learned models on levels ($R^2 = 0.817$)—is the *best* directional ranker (AUC-ROC = 0.792), whereas XGBoost leads on levels but trails ridge on AUC (0.779). The champion XGBoost nonetheless delivers 80.2% overall directional accuracy on the holdout.

The naive baseline posts the highest F1 (0.698) despite mediocre AUC. This is not an artifact but a structural property: in strongly weekly markets, yesterday-sign persistence inherits high base rates of repeated signs, inflating raw hit counts (and hence F1 of the binarized predictor) even when the *ranking* of moves—what AUC measures—is weak. The lesson generalizes: for trading applications, report rank-based skill (AUC) alongside count-based skill (F1/accuracy), and select the deployment model per objective. Level-accuracy champions need not be direction champions.

6 Discussion

Why do calendar and lag features suffice? Gain-based feature importances consistently rank the lag-1d and lag-7d prices, trailing rolling means, and intra-day position (block/hour bucket) at the top, with seasonal and holiday terms contributing secondary structure. The DAM’s uniform-price, day-ahead-closed design implies that the aggregated bid curve embeds all private information—including weather expectations—by gate closure. What remains exploitable publicly is precisely the regularity the bid process itself imposes: daily and weekly cycles, peak/valley shapes, and short-memory autocorrelation. Our ablation (Section 5.2) is a direct empirical confirmation.

Why do boosted trees win? The mapping from features to price is non-linear and interaction-heavy (e.g., evening-peak multiplier $\times$ summer season $\times$ post-holiday weekday). Additive tree ensembles approximate such interactions natively, are robust to the heavy-tailed, ceiling-pinned response distribution, and require minimal preprocessing. Their dominance matches the tabular-learning literature and the European-market evidence reviewed in [Lago et al. \(2021\)](#).

Model-objective matching. The ridge/AUC result (Section 5.4) suggests a practical ensemble: use boosted trees for scheduled-volume planning where level errors dominate cost, and ridge (or a rank-calibrated variant) for directional positioning, or blend signals. Because both train in seconds-to-minutes on this feature matrix, maintaining both is cheap.

Cap effects. Blocks pinning near the Rs10,000/MWh ceiling concentrate the largest residuals. Tree models degrade gracefully there, but a binary “cap-proximity” indicator, or a dedicated spike model layered on the main regressor, is a natural next step.

7 Limitations and Future Work

1. **Point forecasts only.** Trading and risk use cases ultimately require calibrated intervals; conformalized quantile regression or distributional boosting atop the same features is the highest-value extension ([Nowotarski and Weron, 2018](#)).
2. **Sequence models unbonded.** The released LSTM is not yet benchmarked head-to-head under the identical protocol; attention-based multi-horizon architectures ([Lim et al., 2021](#)) may exploit cross-block dependencies that per-block tabular features ignore.
3. **Unused order-book depth.** Purchase bids, sell bids, and cleared volumes are ingested but not yet included in the feature matrix used by the benchmarked models; bid-book imbalance features are a promising lever, particularly for spike regimes.
4. **Two-city weather proxy.** The null weather result applies to Delhi/Mumbai apparent temperature as a national-load proxy; richer aggregates (degree-day indices over multiple regions, hydrology, renewable-output forecasts) were not tested.
5. **Single market.** Findings are specific to IEX DAM mechanics; transfer to other exchanges (e.g., real-time/GDAM segments or European queues) requires re-validation.
6. **No retraining cadence.** Stability was measured for fixed models; rolling refits could tighten errors further, at added operational complexity.

8 Conclusion

We presented a rigorous, reproducible study of day-ahead electricity price forecasting on the Indian Energy Exchange at its native 15-minute resolution, evaluating five model families over a 17-month untouched holdout—a horizon rarely examined in the literature. Gradient-boosted ensembles explain 86.7% of price variance (RMSE 966 Rs/MWh, WAPE 13.8%), cutting naive-baseline error by 43%; calendar and shifted autoregressive features drive essentially all of the achievable accuracy, with city-level weather adding nothing measurable; and directional skill decouples from level skill, rewarding objective-specific model selection. By releasing the full pipeline openly, we aim to provide the Indian power-market community with a transparent baseline against which future methods—probabilistic, sequence-based, and order-book-aware—can be measured honestly.

Reproducibility Statement

All code (scrapers, feature engineering, model definitions, benchmarking, dashboards), exact feature lists, and hyperparameters are available at <https://github.com/uttampaliwal/power-price-predictor> (Paliwal, 2026). Market data are publicly downloadable from the IEX market snapshot portal (Indian Energy Exchange Ltd., 2026); weather histories are available from Open-Meteo (Zippenfenig, 2026). Experiments are seeded and deterministic given library versions pinned in the repository.

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